

BloodSync

A Web-based Blood Donor Coordination and Management System

Project Proposal

Software Dev. Group Project

(ICT2410_UOG04/07)

Group No: 7

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Declaration

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Abstract

The **Blood Donor Coordination and Management System (BloodSync)** is a centralized web-based platform that enhances the efficiency of blood donation coordination by securely connecting donors, hospitals, and blood banks. The system streamlines donor registration, eligibility verification, appointment scheduling, and communication, reducing delays during urgent blood shortages.

Key functional components include Donor Registration and Profile Management, a mandatory Donor Health Pre-Screening module, a prioritized Blood Request Queue System, a restricted-access Live Blood Stock Dashboard for verified medical staff, and donor retention features such as badges and eligibility reminders.

The system focuses strictly on **information management, scheduling, and communication**. Developed using Laravel and PostgreSQL, BloodSync provides a scalable and secure platform designed for academic evaluation and real-world applicability, improving the reliability and responsiveness of blood donation coordination.

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Chapter 1 - Introduction

1.1 Introduction

The **BloodSync – Blood Donor Coordination and Management System** is a web-based platform created to enhance the efficiency, responsiveness, and reliability of blood donation workflows. The system provides a centralized digital environment that connects donors, hospitals, and blood banks through a secure and structured information management framework.

In many regions, the coordination of blood donors is still dependent on manual communication, outdated donor lists, and disconnected hospital processes. These traditional methods often result in delays during emergencies, difficulty in verifying donor eligibility, and poor visibility into blood inventory levels. BloodSync aims to address these challenges by digitizing the donor registration process, automating eligibility checks, facilitating hospital blood requests, and supporting real-time notifications.

By integrating appointment scheduling, donor health pre-screening, and inventory visibility into one cohesive platform, BloodSync significantly improves workflow efficiency for hospitals while promoting continuous donor engagement. This project demonstrates how modern information systems can strengthen healthcare infrastructure and contribute to timely, life-saving medical responses.

1.2 Motivation

The motivation behind BloodSync arises from the observed gaps in traditional blood donation management:

- **Manual donor outreach** often leads to delayed responses during emergencies.
- **Fragmented donor records** make it difficult to track donation history or eligibility.
- **Hospitals lack tools for prioritizing donors** based on urgency, blood type, and proximity.
- **No unified communication platform** exists between hospitals and blood banks for real-time coordination.
- **Donor retention is low** due to lack of follow-up reminders or recognition.

These limitations can directly impact patient outcomes, especially during critical shortages. The need for a unified, digital, and automated platform is essential to modernize and streamline healthcare operations.

BloodSync seeks to overcome these issues by providing:

- Centralized donor information management
- Automated donor eligibility calculation
- A clean communication channel between hospitals and donors
- A structured blood request queue system
- Tools that encourage donor retention and health safety

1.3 Aim and Objectives

Aim:

To develop a secure and efficient web-based coordination platform that connects donors, hospitals, and blood banks to streamline donor management, improve emergency response times, and enhance overall blood donation workflow.

Objectives:

- **Develop a donor profile and registration module** with identity verification and donation history tracking.
- **Implement a health pre-screening form** to identify temporary ineligibility before appointment scheduling.
- **Create a hospital request system** that allows posting, prioritizing, and tracking blood requests.
- **Provide a Live Blood Stock Dashboard** visible only to verified hospital and blood bank staff.
- **Enable appointment scheduling** with available time slots and hospital confirmations.
- **Incorporate donor motivation features** such as non-monetary badges and reminders.
- **Establish real-time alerts and notifications** for urgent blood needs and donor eligibility.
- **Ensure secure access control** by implementing role-based authentication for all user types.

1.4 Scope of the Project

In-Scope Features:

- Donor registration & identity verification
- Donor profile management and donation history tracking
- Automated eligibility calculation using the 4-month rule
- Health pre-screening form for donor safety
- Blood request queue system for hospitals and blood banks
- Priority donor matching based on blood type, location, and eligibility
- Live Blood Stock Dashboard (restricted access)
- Appointment booking with hospital confirmation
- Real-time notifications and reminders
- Role-based access control for Admin, Donor, Hospital User, Lab Technician

Future Enhancements:

- Mobile application for donors and hospitals
- SMS gateway integration for areas with limited internet
- AI-driven donor matching and urgency forecasting
- QR code-based donor check-in
- Multi-language support for broader reach
- Integration with national health databases

Chapter 2 – System Analysis

2.1 Explaining the Current System (Existing Process)

The current blood donation workflow is heavily manual and fragmented. Donor information is often stored in spreadsheets, paper records, or isolated hospital systems that are not interconnected. When blood is urgently needed, hospital staff must manually search for suitable donors, relying on outdated lists, phone calls, and informal communication channels.

This leads to several issues:

- **Slow emergency response** due to difficulty locating eligible donors
- **No automated eligibility tracking**, resulting in contacting ineligible donors
- **Data inconsistencies** between different hospitals or blood banks
- **Lack of a central platform** to manage nationwide or regional donor communication
- **No visibility of blood stock levels** to support informed decision-making

Typical flow in the current process:

1. A hospital identifies the need for a blood type.
2. Staff search existing donor lists manually.
3. Donors are contacted one by one, often without immediate response.
4. Donation schedules must be coordinated manually.
5. Blood stock inventory is updated manually after collection.

These delays and inefficiencies highlight the urgent need for an automated, centralized, and data-driven system.

2.2 Review of Similar Systems

(a) Red Cross Blood Donation Portal (International)

Strong donor management and appointment scheduling system.
However, it lacks localized donor prioritization, real-time hospital request handling, and integrated blood inventory visibility.

(b) Sri Lanka National Blood Transfusion Service (NBTS) System

Maintains national donor data and organizes mobile campaigns.
But the system is **not designed for real-time donor–hospital coordination** and has limited public-facing digital features.

(c) BloodNest Mobile Application (India)

A peer-to-peer donor–recipient app.
It lacks **verification mechanisms, hospital integration, and structured donor eligibility checks**.

Table 2.1: Summary of Limitations in Existing Systems

System	Strengths	Limitations
Red Cross Portal	Reliable donor scheduling	No donor prioritization or hospital request pipeline
NBTS SL	National database	No real-time donor coordination
BloodNest App	Fast connections	No hospital verification, no inventory system

BloodSync addresses all these limitations by integrating donor matching, request queues, eligibility tracking, and inventory monitoring into a single system.

2.3 Identifying and Prioritizing Requirements

The system requirements are categorized into **functional** and **non-functional** aspects to ensure a comprehensive understanding of user needs and performance expectations.

2.3.1 Functional Requirements

ID	Requirement	Description
FR1	Donor Registration	Manage donor profiles and identity verification
FR2	Health Pre-Screening	Collect data for temporary eligibility checks
FR3	Donor Eligibility Tracking	Apply 4-month rule automatically
FR4	Blood Request Management	Hospitals can post and track requests
FR5	Priority Matching	Match eligible donors by type, proximity, and urgency
FR6	Appointment Booking	Donors schedule donation time slots
FR7	Inventory Management	Lab technicians update processed units
FR8	Live Stock Dashboard	Hospitals view real-time stock levels
FR9	Notifications	Alerts for donors and hospitals
FR10	Admin Panel	Full system oversight and user management

2.3.2 Non-Functional Requirements

ID	Requirement	Description
NFR1	Performance	The system must respond to user actions within 2 seconds under normal load.
NFR2	Scalability	Should handle 10,000+ concurrent users efficiently.
NFR3	Security	Role-based authentication, encrypted storage.
NFR4	Availability	Ensure at least 99.5% system uptime.
NFR5	Usability	Interface should be accessible, intuitive, and mobile-friendly.

ID	Requirement	Description
NFR6	Reliability	Include backup and recovery mechanisms.
NFR7	Maintainability	Follow MVC architecture for easy maintenance and updates.

2.4 Selection of Process Model and Justification

The **Agile Development Model** is chosen due to:

- Iterative and incremental development
- Flexibility to evolve donor screening and request workflows
- Early availability of prototypes for hospital feedback
- Continuous improvement of matching algorithms
- Reduced risk through frequent testing
- Efficient collaboration with stakeholders

Agile ensures the system remains user-oriented, scalable, and adaptable to healthcare needs.

Chapter 3 – System Design

3.1 Introduction

This chapter outlines the technical design of BloodSync, including architecture, UML diagrams, interface designs, and development timeline. The design ensures that the system is robust, secure, and optimized for multi-user performance.

3.2 System Architecture

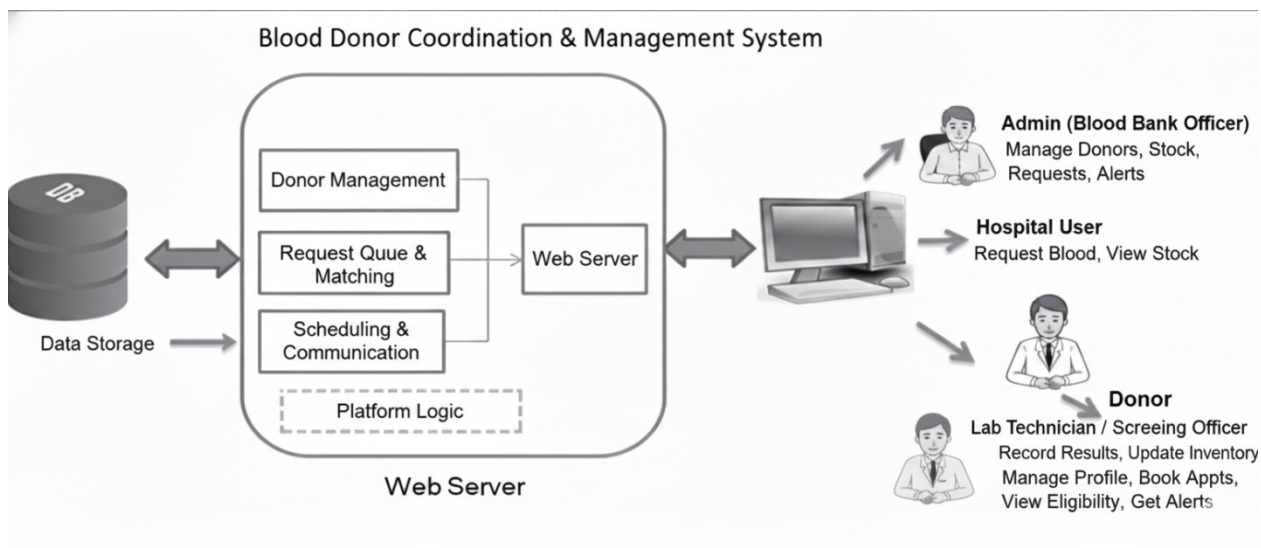


Figure 1: Blood Donor Coordination and Management System Architecture

The architecture consists of the following interconnected components:

- **Client Interface:** Represents the user's access point(Web browser/mobile interface) to the system, connecting to the Web Server.
- **Web Server (Application Core):** Hosts the Laravel application, which handles the core business logic.
 - **Controller:** Receives user requests from the Client, delegates processing tasks to the Model, and selects the appropriate View to render the result.
 - **Model:** Encapsulates the application's data and business logic. The diagram specifies key modules: **Donor**, **Staff**, and **Inventory**. It manages all interactions with the database.

- **DB (Database):** The persistent storage layer, implemented with ****PostgreSQL****. It holds all critical data including user profiles, appointment schedules, blood stock records, and donation history.
- **User Roles:** The primary actors interacting with the system include the **Administrator**, **Donor/Patient** (submitting requests), **Medical Staff** (hospital users), and **Lab Technician / Screening Officer**.

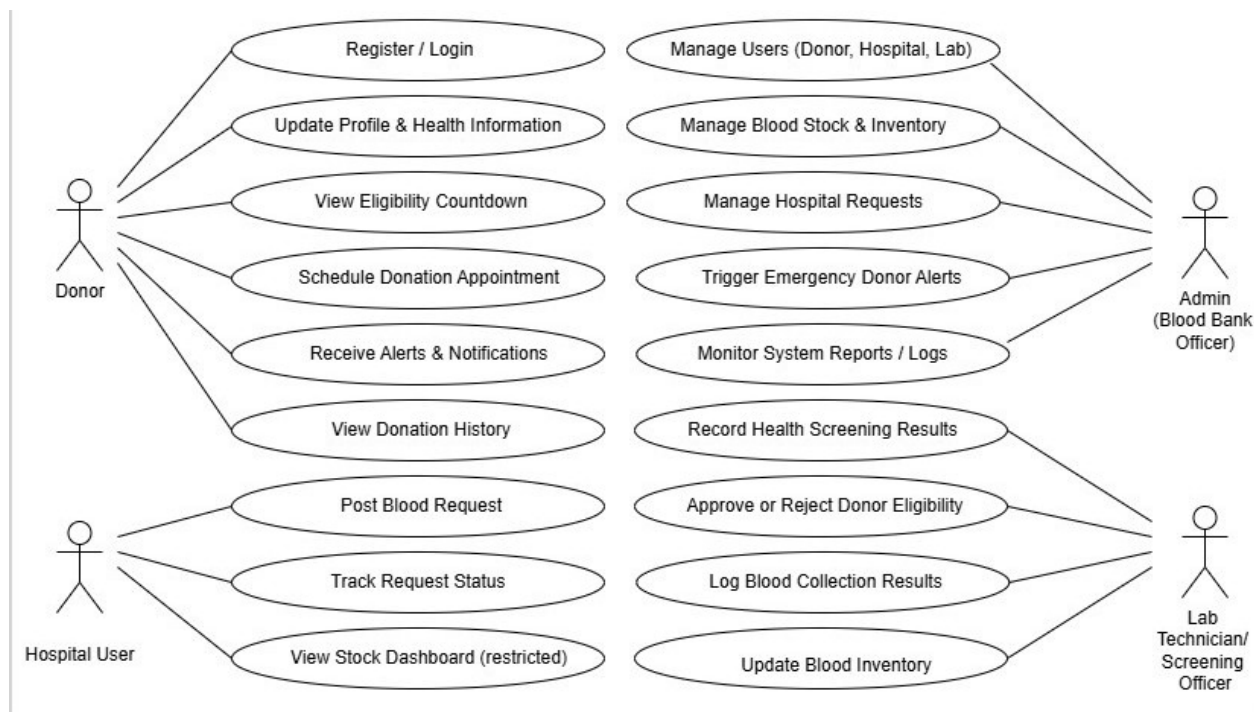
This centralized architecture ensures efficient data flow, streamlined processing, and provides a secure, single source of truth for all coordination activities.

3.3 UML Diagrams

3.3.1 Use Case Diagram

The Use Case Diagram visually represents the system's boundary and the primary functional interactions available to the four defined user roles: Donor, Hospital User, Admin (Blood Bank Officer), and Lab Technician/Screening Officer. Key use cases include Manage Profile, Complete Pre-Screening, Book Appointment, Post Blood Request, View Live Stock, and Manage Donor/Stock.

Figure 3.1: Use Case Diagram of BDMS



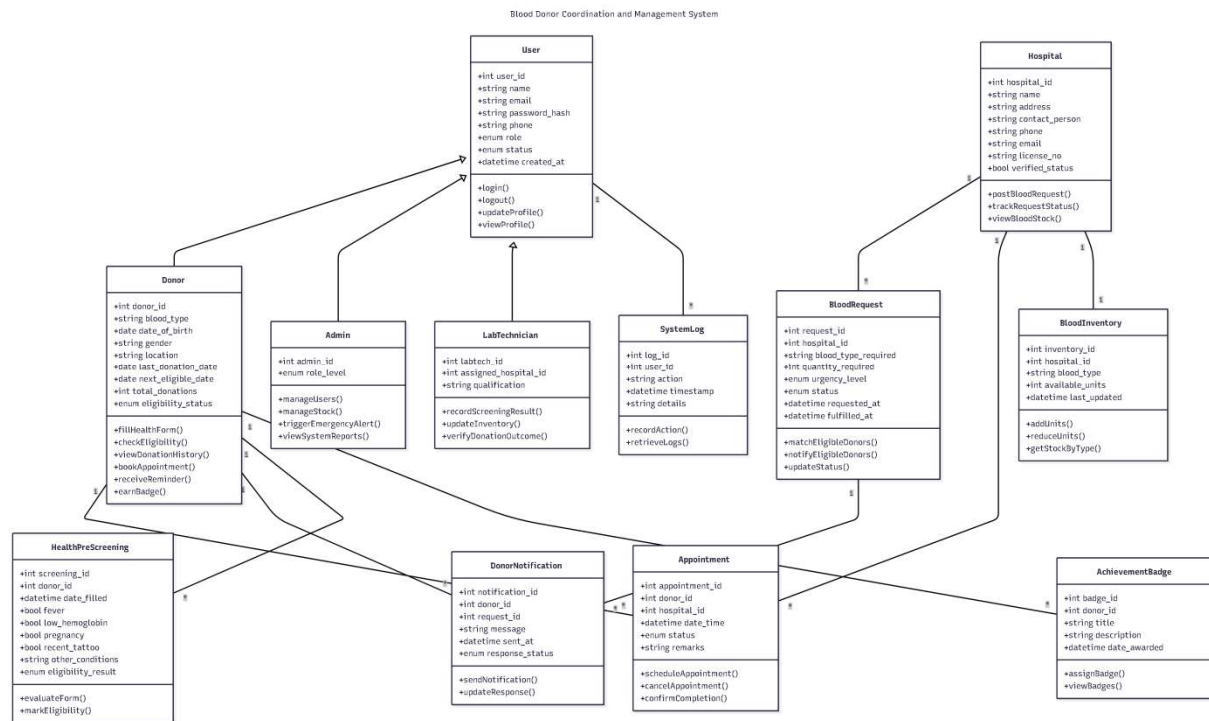
3.3.2 Class Diagram

The Class Diagram illustrates the structural relationships between entities in the BloodSync system.

Key classes include:

- **User**
(attributes: *user_id*, *name*, *email*, *role*, *contact*, *login_credentials*)
- **Donor**
(*donor_id*, *blood_type*, *last_donation_date*, *next_eligible_date*, *donation_history*)
- **Hospital**
(*hospital_id*, *hospital_name*, *location*, *verification_status*)
- **BloodRequest**
(*request_id*, *hospital_id*, *blood_type_required*, *urgency_level*, *request_status*)
- **Appointment**
(*appointment_id*, *donor_id*, *hospital_id*, *appointment_date*, *time_slot*, *status*)
- **HealthPreScreening**
(*screening_id*, *donor_id*, *screening_responses*, *eligibility_status*)
- **BloodInventory**
(*inventory_id*, *blood_type*, *unit_count*, *last_updated_date*)
- **Admin**
(*admin_id*, *system_control_functions*)

Figure 3.2: Class Diagram of BDMS



3.4 User Interface Design and Wireframes

The User Interface (UI) of BloodSync is designed with a strong focus on simplicity, clarity, and usability to support all key stakeholders, including donors, hospital users, lab technicians, and administrators. The layout follows a clean and responsive design approach to ensure smooth accessibility across desktop and mobile devices.

3.4.1 Design Considerations

- **Simplicity:**
Interfaces use clear navigation menus, minimal clutter, and intuitive layouts to help users perform tasks quickly.
- **Consistency:**
Uniform color schemes, icons, and typography are used throughout the system to create a seamless user experience.
- **Accessibility:**
Adequate contrast, readable text sizes, and responsive page structures ensure usability for users with varying device types and abilities.
- **Role-Based Interfaces:**
Each user type—Donor, Hospital User, Lab Technician, and Admin—has a customized dashboard displaying only the features relevant to their responsibilities.

3.4.2 Wireframe Overview

Main Screens:

1. **Home Page:**
Provides an overview of the system with information such as donation guidelines, platform features, and options to log in or register as a donor or hospital user
2. **Donor Dashboard:**
Displays donor profile details, last donation date, next eligible donation date, donation history, eligibility status, and appointment booking options.
3. **Hospital Dashboard:**
Provides tools for posting blood requests, viewing request status, accessing the Live Blood Stock Dashboard, and managing hospital-specific settings.
4. **Blood Request Management Page:**
Allows hospital users to submit new blood requests and track existing ones through a detailed request list with urgency levels and statuses.
5. **Lab Technician Panel:**
Enables lab technicians to record screening results, update donation outcomes, and manage inventory updates for processed blood units.
6. **Live Blood Stock Dashboard:**
Displays current blood unit counts by blood type, accessible only to authorized hospital staff and administrators.

7. Admin Panel:

Offers complete system oversight including user management, role assignment, configuration settings, and monitoring system-wide activities.

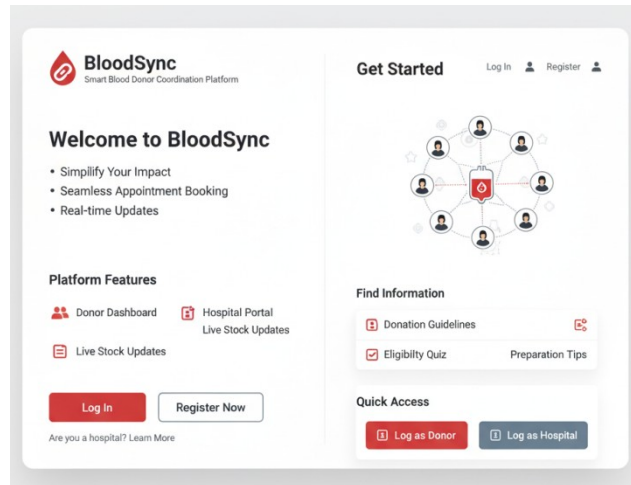


Figure 1: Home Page

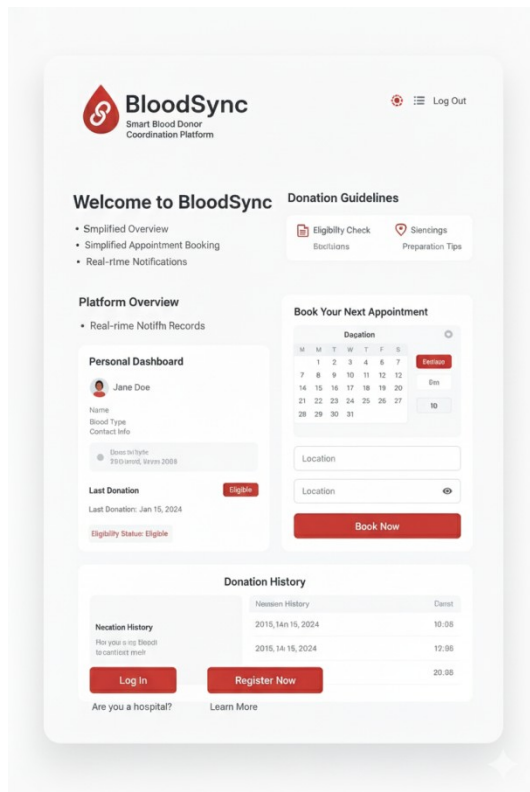


Figure 2: Donor Dashboard

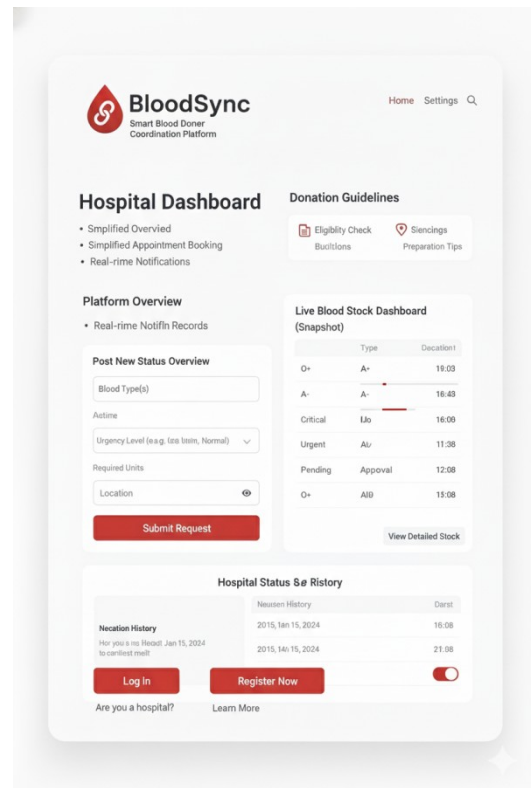


Figure 3: Hospital Dashboard

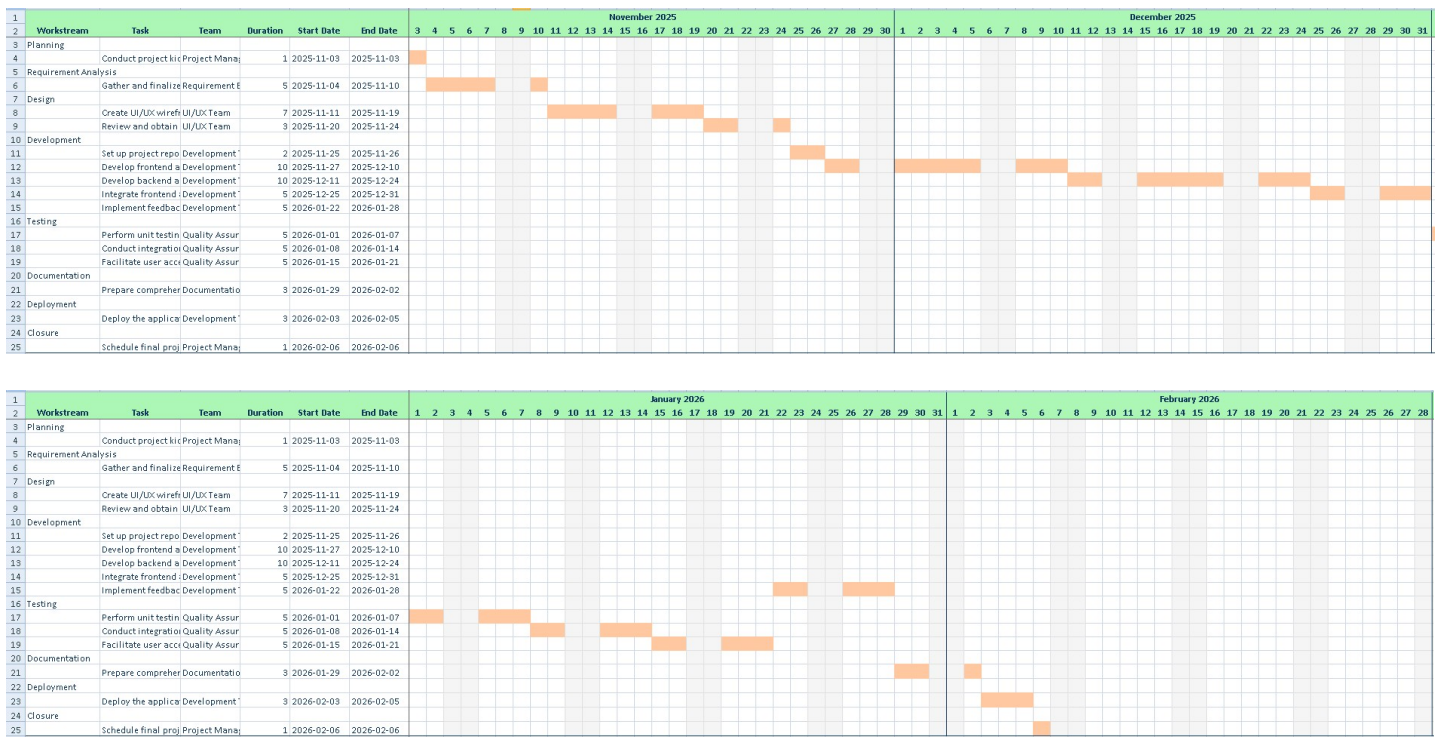


3.5 Project Timeline (Gantt Chart)

The following Gantt chart outlines the major phases and tasks of the project, along with their respective durations.

Phase	Tasks	Duration
Week 1	Requirement analysis & feasibility study	1 week
Week 2	System design (architecture, ERD, UI planning)	1 week
Weeks 3–5	Backend & frontend development (Laravel MVC)	3 weeks
Weeks 6–7	Database setup, integration, and unit testing	2 weeks
Week 8	System testing & debugging	1 week
Week 9	User Acceptance Testing (UAT)	1 week
Week 10	Final documentation & deployment	1 week

Figure 3.10: Gantt Chart of Project Schedule



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